TOPIC 4

PARALLEL PLATE WAVEGUIDES

4.2.6 Cut-off equations for TE and TM modes

We need to find the characteristics of TE and TM modes in a parallel plate waveguide. We know $h^2 = \gamma^2 + \omega^2\mu\varepsilon$, where h = mπ/a. Hence, the propagation constant is

$$\gamma = \sqrt{\left(\frac{m\pi}{a}\right)^2 - \omega^2\mu\varepsilon} \qquad (4.26)$$

For propagation to take place γ should be **imaginary** and hence $\left(\frac{m\pi}{a}\right)^2 < \omega^2\mu\varepsilon$, i.e. $\gamma = ik$, and rearranging eq. 4.26 in this sense gives

$$\gamma = \sqrt{-\left[\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2\right]} = i\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2} = 0 + ik = \alpha + ik \text{ or}$$

Phase constant is $k = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2}$ (4.27a)

Attenuation $\alpha = 0$ (4.27b)

Using the condition $\omega_c^2\mu\varepsilon = \left(\frac{m\pi}{a}\right)^2$ we get cut-off expressions

$$f_c = \frac{m}{2a\sqrt{\mu\varepsilon}} \qquad (4.28)$$

The guide wavelength is given by

$$\lambda_g = \frac{2\pi}{k} = \frac{2\pi}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2}} \qquad (4.29)$$

The phase velocity is given by

$$v_p = \frac{\omega}{k} = \frac{\omega}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2}} \qquad (4.30)$$

4.2.7 Expressions for TEM modes

We now need to find the characteristics of TEM mode in a parallel plate waveguide.

This is a special case of TM mode when $m = 0$. Hence the field components are found by putting m = 0 in eqs. 4.23, 4.24 and 4.25. That is

$$H_z = 0$$

$$H_y = C_4 \cos\left(\frac{m\pi x}{a}\right) e^{-ikz} = C_4 e^{-ikz} \tag{4.31}$$

$$E_x = \frac{\gamma C_4}{i\omega\varepsilon} \cos\left(\frac{m\pi x}{a}\right) e^{-ikz} = \frac{k}{\omega\varepsilon} C_4 e^{-ikz} \tag{4.24}$$

$$E_z = \frac{-C_4 m\pi}{i\omega\varepsilon a} \sin\left(\frac{m\pi x}{a}\right) e^{-ikz} = 0 \tag{4.25}$$

4.2.8 Cut-off expressions for TEM mode

Letting m=0

Propagation constant $\gamma = i\beta = i\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2} = i\sqrt{\omega^2\mu\varepsilon - 0} = \alpha + ik = i\omega\sqrt{\mu\varepsilon}$ (4.34)

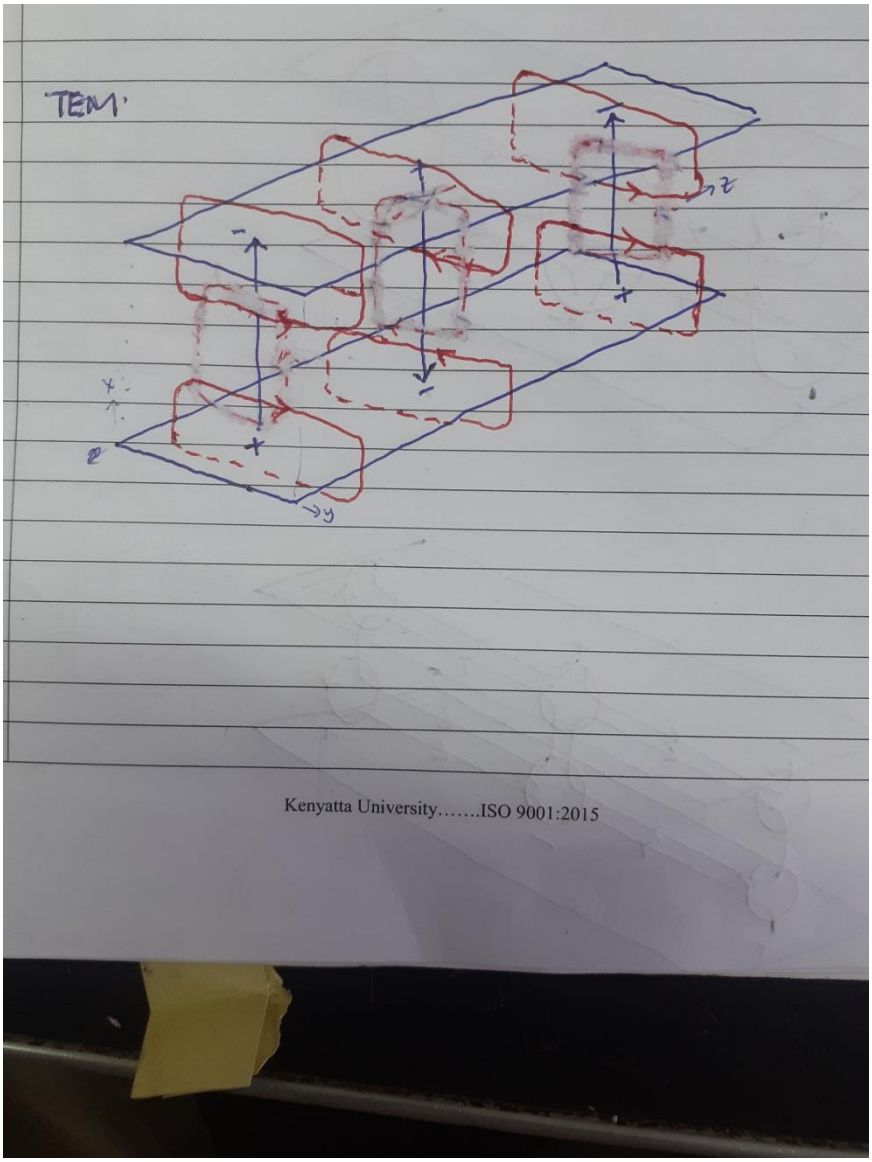
Attenuation α = 0

Phase constant $k = \omega\sqrt{\mu\varepsilon}$ (condition for TEM!) (4.35)

Phase velocity $v_p = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu_o\varepsilon_o}} = 3 \times 10^8\ m/s$ (4.36)

Guide wavelength $\lambda_g = \frac{2\pi}{k} = \frac{2\pi}{\sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2}} = \frac{2\pi}{\omega\sqrt{\mu\varepsilon}} = \lambda_o$ (4.37)

[3]



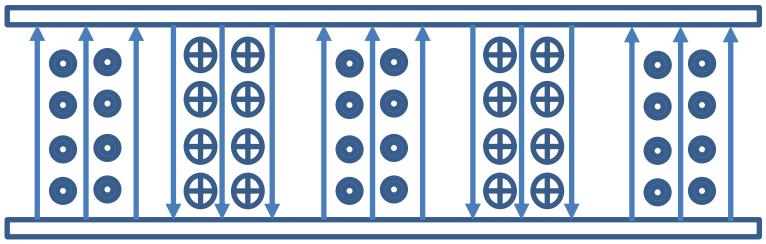


Figure 4.4 Field configuration for TEM mode

4.2.9 Power flow in a parallel plate waveguide

The power flow in a parallel plate waveguide, is given by the product of power density and cross-sectional area.

$$P_T = SA = A\left[\vec{E} \times \vec{H}\right] = A\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ E_x & E_y & E_z \\ H_x & H_y & H_z \end{vmatrix} \tag{1.56}$$

$$S = \left(E_y H_z - H_y E_z\right)\hat{x} + \left(E_z H_x - E_x H_z\right)\hat{y} + \left(E_x H_y - H_x E_y\right)\hat{z} \tag{1.57}$$

Since the propagation is in the z-direction the x and y components are ignored

$$P_T = SA = \frac{1}{2}\iint \left[E_x H_y - H_x E_y\right] dxdy \tag{1.58}$$

Substituting for $E_y = -\eta H_x$ and $E_x = \eta H_y$

$$P_T = \frac{\eta}{2}\iint \left[H_y^2 + H_x^2\right] dxdy \tag{1.58b}$$

Substituting for $H_x = -E_y/\eta$ and $H_y = E_x/\eta$

$$P_T = \frac{1}{2\eta}\iint \left[E_y^2 + E_x^2\right] dxdy \tag{1.58c}$$

4.2.10 Power flow in a parallel plate waveguide for TE and TM modes

$$P_T = SA = \frac{1}{2}\iint \left[E_x H_y - H_x E_y\right] dxdy = \frac{\eta}{2}\iint \left[H_y^2 + H_x^2\right] dxdy$$

$E_z = H_y = E_x = 0$

$$H_x = \frac{-\gamma C_1}{i\omega\mu}\sin\left(\frac{m\pi x}{a}\right)e^{-i\beta z}$$

$$E_y = C_1 \sin\left(\frac{m\pi x}{a}\right)e^{-i\beta z}$$

$$H_z = \frac{-mx}{i\omega\mu a} C_1 \cos\left(\frac{m\pi x}{a}\right)e^{-i\beta z}$$

Example

Find the power propagated in TE_1 mode in a parallel plate waveguide

$$P_T = \frac{\eta}{2}\iint \left[H_y^2 + H_x^2\right] dxdy$$

Field components where m=1

$$E_z = 0$$

$$H_x = \frac{-\gamma C_1}{i\omega\mu}\sin\left(\frac{m\pi x}{a}\right)e^{-i\beta z} = \frac{-\gamma C_1}{i\omega\mu}\sin\left(\frac{\pi x}{a}\right)e^{-i\beta z} \quad (4.14)$$

$$E_y = C_1\sin\left(\frac{m\pi x}{a}\right)e^{-i\beta z} = C_1\sin\left(\frac{\pi x}{a}\right)e^{-i\beta z} \quad (4.15)$$

$$H_z = \frac{-mx}{i\omega\mu a}C_1\cos\left(\frac{m\pi x}{a}\right)e^{-i\beta z} = \frac{-x}{i\omega\mu a}C_1\cos\left(\frac{\pi x}{a}\right)e^{-i\beta z} \quad (4.16)$$

$H_y = 0$

Substituting H_y and H_x, gives

$$P_T = \frac{\eta}{2}\iint \left[H_x^2\right] dxdy = \frac{\eta}{2}\int_0^a \left(\frac{-\gamma C_1}{i\omega\mu}\right)^2 sin^2\left(\frac{\pi x}{a}\right) dx \int_0^d dy$$

Using $sin^2\theta = \frac{1}{2}(1 - cos2\theta)$

$$P_T = \frac{\eta}{2}\iint \left[H_x^2\right] dxdy = \frac{\eta}{2}\int_0^a \left(\frac{-\gamma C_1}{i\omega\mu}\right)^2 \frac{1}{2}\left(1 - cos\left(\frac{2\pi x}{a}\right)\right) dx \int_0^d dy$$

$$P_T = \frac{\eta}{2}\left(\frac{-\gamma C_1}{i\omega\mu}\right)^2 \frac{a}{\pi}\frac{ad}{2} = \left(\frac{-\gamma C_1}{i\omega\mu}\right)^2 \frac{a^2 d\eta}{4\pi}$$

Example

Find the power propagated in TM_1 mode in a parallel plate waveguide

$$P_T = SA = \frac{1}{2\eta}\iint \left[H_y^2\right] dxdy$$

$H_z = H_x = E_y = 0$

$$H_y = C_4\cos\left(\frac{m\pi x}{a}\right)e^{-ikz} = C_4\cos\left(\frac{\pi x}{a}\right)e^{-ikz}$$

$$E_x = \frac{\gamma C_4}{i\omega\varepsilon}\cos\left(\frac{m\pi x}{a}\right)e^{-ikz} = \frac{\gamma C_4}{i\omega\varepsilon}\cos\left(\frac{\pi x}{a}\right)e^{-ikz}$$

$$E_z = \frac{-C_4 m\pi}{i\omega\varepsilon a}\sin\left(\frac{m\pi x}{a}\right)e^{-ikz} = \frac{-C_4 m\pi}{i\omega\varepsilon a}\sin\left(\frac{\pi x}{a}\right)e^{-ikz}$$

$$P_T = \frac{\eta}{2}\int_0^a (C_4)^2\, cos^2\left(\frac{m\pi x}{a}\right)dx\int_0^d dy = (C_4)^2\frac{a^2 d\eta}{4\pi}$$

4.2.11 Power flow in a parallel plate waveguide for TEM mode

$$P_T = SA = \frac{1}{2}\iint \left[E_x H_y - H_x E_y\right]dxdy = \frac{\eta}{2}\iint \left[H_y^2 + H_x^2\right]dxdy$$

Field components are

$$H_y = C_4 e^{-ikz} \tag{4.31}$$

$$E_x = \frac{\beta}{\omega\varepsilon}C_4 e^{-ikz} \tag{4.32}$$

$$E_z = 0$$

$H_x = 0$

Example
Find the power propagated in TEM mode in a parallel plate waveguide

$$P_T = SA = \frac{1}{2}\iint \left[E_x H_y - H_x E_y\right]dxdy = \frac{\eta}{2}\iint \left[H_y^2 + H_x^2\right]dxdy$$

$$H_y = C_4 e^{-ikz} \tag{4.31}$$

$$E_x = \frac{\beta}{\omega\varepsilon}C_4 e^{-ikz} \tag{4.32}$$

$$E_z = 0$$

$H_x = 0$

$$P_T = \frac{\eta}{2}\iint \left[H_y^2\right]dxdy = \frac{\eta}{2}\iint (C_4)^2\, dxdy = \frac{\eta a d}{2}(C_4)^2$$

Example 4.1
A parallel-plate waveguide having plate separation of 1 cm is filled with Teflon ($\varepsilon_r = 2.08$). The waveguide is oriented along the z-direction. The center conductor of a coaxial cable is extended into space between the plates in the z-direction, exciting a mode(s) propagating in the z-direction with field components $E_z \neq 0$. The source does not

excite TEM mode. Find the (a) propagating mode(s) (b) phase constant (c) usable frequency range for single-mode propagation (d) what happens at sufficiently high frequencies.

Solution

(a) $E_z \neq 0$, implies TM modes.

(b)

$$\beta = \sqrt{k^2 - \left(\frac{n\pi}{d}\right)^2} = \sqrt{2.08\omega^2\mu_o\varepsilon_o - (100n\pi)^2}$$

(c)

n = 0 is TEM mode
n = 1 is first TM mode

$$f_{c1} = \frac{m}{2a\sqrt{\mu\varepsilon}} = \frac{mc}{2a\sqrt{\mu\varepsilon}} = \frac{1\times3\times10^8}{2\times0.01\sqrt{2.08\times1}} = 10.4\ GHz$$

n = 2 is second TM mode

$$f_{c2} = \frac{m}{2a\sqrt{\mu\varepsilon}} = \frac{mc}{2a\sqrt{\mu\varepsilon}} = \frac{2\times3\times10^8}{2\times0.01\sqrt{2.08\times1}} = 20.8\ GHz$$

Thus, the usable frequency range is from 10.4 GHz to 20.8 GHz.

4.3.3 Assignment

1 The cutoff frequency of the TE_2 and TM_2 modes in a parallel-plate waveguide is 100 GHz. If d= 1mm, determine the ε_r of the medium between the plates. [9]

2 A parallel plate waveguide is to be designed so that only TEM modes can propagate in the frequency range $0 < f < 2$ GHz. The ε_r and μ_r are 9 and 1, respectively. (a) What is the maximum allowed spacing between the parallel plate waveguides plates? (b) If the plate spacing is 2.1 cm, and f = 10 GHz, what TE_n and TM_n modes will propagate? [0.5 cm, TE_n, TM_n n = 1,2,3,4]

3 A parallel-plate waveguide with a = 2 mm is partially filled with two dielectrics with $\varepsilon_{r1} = 4$ and $\varepsilon_{r2} = 2.1$. At a certain frequency, it is that the TM_1 mode propagates through the guide without suffering any reflective loss at the dielectric interface. Calculate this frequency.[63.9 GHz]

4.4 References

1. David K. Cheng, *Field and Wave Electromagnetic*, Addison-Wesley Pub. Co., New York, 1989.
2. David M. Pozar, *Microwave Engineering*, John Wiley & Sons, Inc., New Jersey, 2005.
3. fotonica.intec.ugent.be

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